

Wahid Al Azad Navid

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Research Interests

Computational genomics: algorithms and probabilistic models for sequence analysis; machine learning for biological sequence data; multimodal learning and its interpretability.

Education

- B.Sc.** **Bangladesh University of Engineering and Technology (BUET)**, Computer Science and Engineering Dhaka, Bangladesh
Jan 2022 – June 2026
- **CGPA: 3.96 / 4.00.** Dean's List in all years.
 - Thesis: *kRISP-meR: A Reference-free Guide-RNA Design Tool for CRISPR/Cas9*. Supervisor: [Dr. Atif Hasan Rahman](#).
- HSC** **Notre Dame College**, Science Dhaka, Bangladesh
Aug 2018 – Jan 2021
- GPA: 5.00 / 5.00

Papers Under Review and Preprints

- A Multimodal Framework for Pavement Condition Assessment Using Roadway Imagery and Vehicle Response** July 2026
 Shams E Shifat, **Wahid Al Azad Navid**, Ashrafur Rahman, AQM Zohuruzzaman, Zhao Gao, Nasif Shafi, Mosharaf Chowdhury, Sadik Khan
 (Under review, [Transportation Research Board \(TRB\) Annual Meeting](#))
- Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization** 2026
 H. M. Shadman Tabib, Istiak Ahmmed Rifi, ..., **Wahid Al Azad Navid**, ... (27 authors)
arxiv.org/abs/2602.14291 (arXiv:2602.14291)

Manuscripts in Preparation

- kRISP-meR: A Reference-free Guide-RNA Design Tool for CRISPR/Cas9** 2026
Wahid Al Azad Navid, Fuad Ahmed Udoy, Mahmudur Rahman Hera, Atif Hasan Rahman
 (Benchmarking complete; manuscript being finalised)
- Disentangling Cross-Modal Information Pathways via Mechanistic Interpretability of Multimodal Fusion Models** 2026
 Abrar Rahman Abir, **Wahid Al Azad Navid**, Sheikh Azizul Hakim
- DocTalkBN: A Novel Dataset of Expert Telemedicine Conversations in Bengali** 2026
 Anik Saha, Mst. Fahmida Sultana Naznin, Sadatul Islam Sadi, Ananya Shahrin Promi, **Wahid Al Azad Navid**, Rifat Shahriyar

Research Experience

- Undergraduate Thesis, BUET CSE, kRISP-meR:** reference-free CRISPR/Cas9 guide-RNA design Dhaka, Bangladesh
May 2025 – June 2026
 Supervised by [Dr. Atif Hasan Rahman](#), in collaboration with [Dr. Mahmudur Rahman Hera](#).
- Designs CRISPR/Cas9 guide RNAs directly from the raw sequencing reads of the individual being edited, without a reference genome or a genome assembly.
 - Models the k-mer frequency spectrum of the reads as a Poisson mixture fitted with expectation-maximization to infer genomic copy number, and from it the expected number of cuts of every candidate guide.

- Benchmarked against CRISPOR and GuideScan2 on human, mouse, and yeast data; rejects guides that appear unique in the reference genome but are duplicated in the individual (CHM13 vs. GRCh38, N22 rice vs. Nipponbare).

LimbicsAI, Applied ML and sensing for infrastructure monitoring (part-time)

Remote

May 2025 – June 2026

- Co-developed a multimodal pavement assessment model fusing roadway video, vehicle-mounted IMUs, and GPS (paper under review at TRB), and built the automated AWS pipeline that processes the field data end to end.
- Built a model on InSAR ground-deformation data; implemented PSI and PCI pavement condition scoring and integrated it into the product.
- Set up the LoRaWAN sensor data pipeline (ChirpStack, EMQX, InfluxDB) and a bridge scour monitoring test system, including the physical sensor setup.
- Wrote crawlers for roadway data collection; rebuilt the company web front end, integrated an AI co-pilot, and deployed the internal documentation server.

Teaching

Presidency University, Lecturer, Department of Computer Science and Engineering

Dhaka, Bangladesh

June 2026 – present

- **Fall 2026:** Object Oriented Programming (theory and lab); Microprocessors and Embedded Systems.
- **Summer 2026:** Structured Programming Language (theory and lab); Digital Logic Design Lab.

Honors and Service

- **Dean's List**, BUET (2022 – 2026)
- **University Merit Scholarship** and **BUET Technical Scholarship** (2022 – 2026)
- **Best UI/UX Award**, Hackathon, BUET CSE Fest (2024)
- **Runners-up**, Project Showcasing, IUT Skill Innovation Fest (2024)
- Government General Scholarship: HSC (2021), SSC (2018), JSC (2016)
- Regional Winner, Bangladesh Junior Science Olympiad (2018)
- Logistics Director, BUET Cyber Security Club (2023 – present); regular capture-the-flag participant

Selected Projects

GradPilot

July 2025

- ML-assisted platform for graduate applications (university matching, SOP review) built as microservices with CI/CD. Spring Boot, Next.js, MySQL, Docker, GitHub Actions.

DHCP Starvation and Spoofing Attack

June 2025

- Simulated both attacks in an isolated lab, measured network disruption, and evaluated DHCP snooping and port security. Python, Scapy, ns-3.

4-bit MIPS Processor

Feb 2024

- Designed the datapath and control unit for a custom instruction set and built it on physical ICs with an ATmega32A.

C Compiler

Dec 2023

- Compiler for a subset of C with scoped symbol tables, type checking, and code generation. Flex, Bison.

Technical Skills

Programming: Python, C/C++, Java

ML and Bioinformatics: PyTorch, NumPy, Pandas, Matplotlib; Jellyfish, Bowtie2, Pilon

Systems and Cloud: AWS, Docker, Git, xv6, ns-3, LoRaWAN (ChirpStack, EMQX, InfluxDB)

Databases and Web: MySQL, PostgreSQL, MongoDB; Node.js, React, Next.js, Spring Boot